ECE 6410 HW02

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Question 1

Question Statement

Use Python to demonstrate one (your choice) AM Modulators discussed in class. You may choose either the Square Law or Balanced Modulator.

Let m(t) be a band-limited 1kHz square wave. You may either synthesize using Fourier Series terms or digital filter (2-pole minimum). Let fc=20kHz and fs=100kHz. Present the time- and frequency-domains of the DSB-SC outpus.

In Python, I used a balanced modulator on a 1kHz square wave. A balanced modulator was selected over the square law because the implementation appeared to be more clean and simple from my understanding. Below is the Python code and plots that demonstrated a successful implementation.

Generated Plot

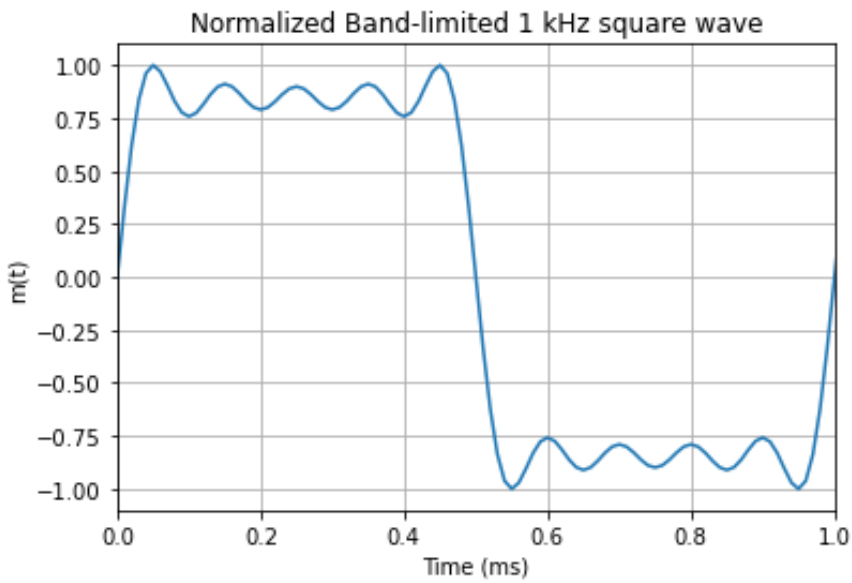

Figure 1: Single period of the generated square wave.

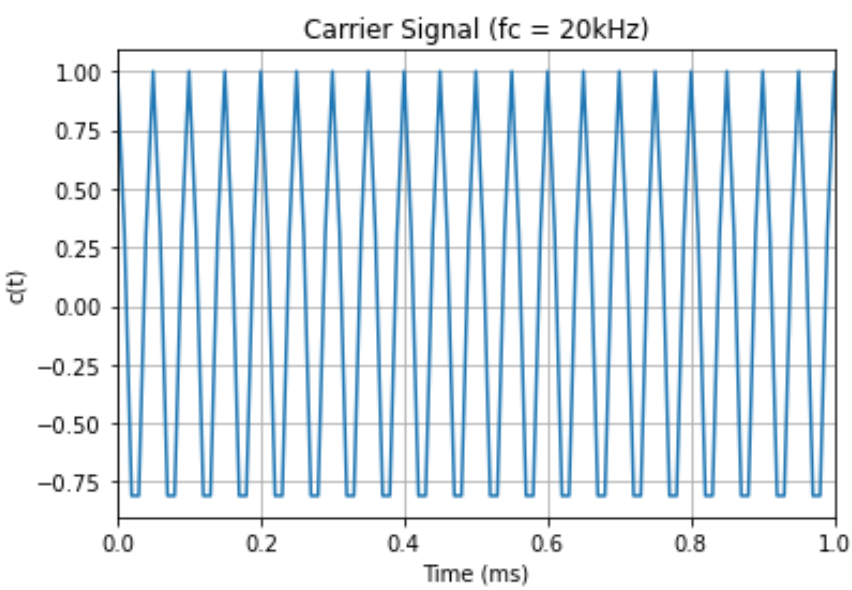

Figure 2: Generated carrier signal.

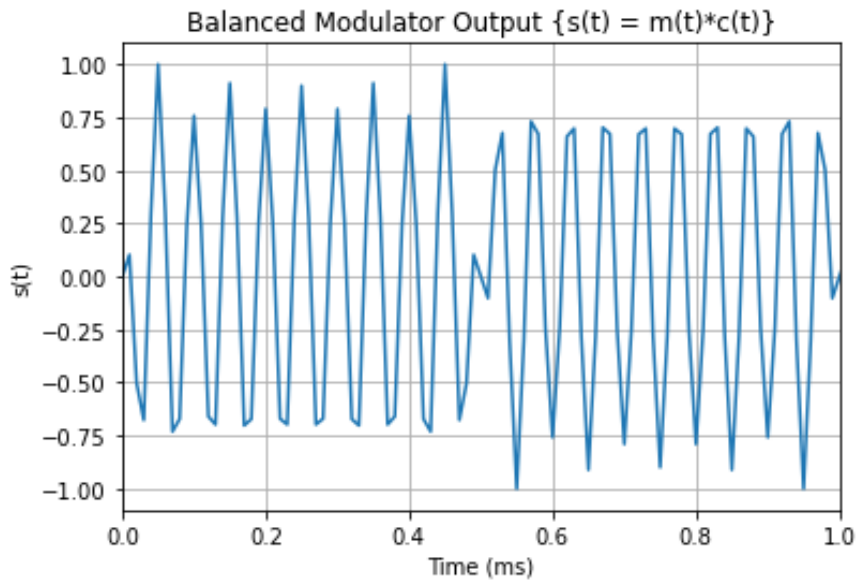

Figure 3: DSB-SC time-domain output from the balanced modulator.

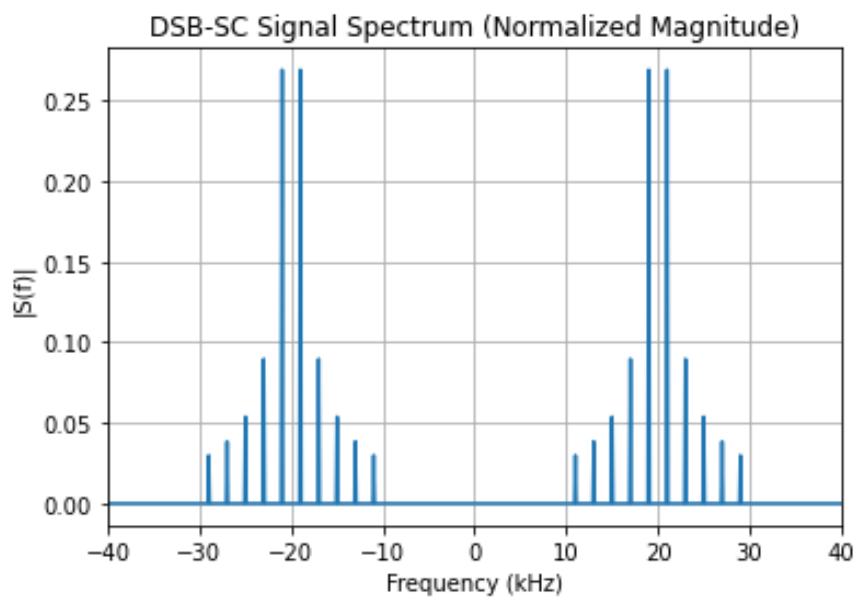

Figure 4: DSB-SC frequency-domain output from the balance modulator. This is normalized magnitude values.

Code

Below is the code used to generate the plots shown above.

```
def p1():
    fc = 20e3                    # hertz
    fs = 100e3                   # hertz
    fm = 1e3                     # message frequency
    T = 1/fm                     # period
```

```python
    T10 = T*10                    # go for 10 periods
    t = np.arange(0, T10, 1/fs)

    # band-limited square wave
    m = np.zeros_like(t)          # array of 0's size of t
    K = 5                         # 5 odd harmonics; 1,3,5,7,9

    # k = 1,3,5,7,9
    for k in range(1, 2*K, 2):
        # equation for square wave build from Fourier series
        m += (4/(np.pi*k))*np.sin(2*np.pi*k*fm*t)

    # normalize square wave
    m = m / np.max(np.abs(m))


    plt.figure()
    plt.plot(t*1000, m)
    plt.xlim(0, 1)      # show 3 ms (about 3 periods of a 1 kHz wave)
    plt.xlabel("Time (ms)")
    plt.ylabel("m(t)")
    plt.title("Normalized Band-limited 1 kHz square wave")
    plt.grid()
    plt.show()

    # create carrier signal
    c = np.cos(2*np.pi*fc*t)

    # plot carrier
    plt.figure()
    plt.plot(t*1000, c)
    plt.xlim(0, 1)      # show 3 ms (about 3 periods of a 1 kHz wave)
    plt.xlabel("Time (ms)")
    plt.ylabel("c(t)")
    plt.title("Carrier Signal (fc = 20kHz)")
    plt.grid()
    plt.show()

    # implement balanced modulator
    s = m * c

    # plot time domain
    plt.figure()
    plt.plot(t*1000, s)
    plt.xlim(0, 1)    # zoom in (carrier is fast)
    plt.xlabel("Time (ms)")
    plt.ylabel("s(t)")
    plt.title("Balanced Modulator Output {s(t) = m(t)*c(t)}")
    plt.grid()
    plt.show()

    # compute FFT
    N = len(s)           # number of samples
    # compute FFT and shift zero frequency to center
```

```python
    s_freq = np.fft.fftshift(np.fft.fft(s))
    # corresponding frequency vector
    frequency = np.fft.fftshift(np.fft.fftfreq(N, 1/fs))

    # plot in freqency domain
    plt.figure()
    plt.plot(frequency/1000, s_freq)            # divide by N to normalize
                                                # frequency/1000 easier plot
    plt.xlim(-40, 40)                           # -40 to 40 kHz
    plt.xlabel("Frequency (kHz)")
    plt.ylabel("S(f)")
    plt.title("DSB-SC Signal Spectrum")
    plt.grid()
    plt.show()

    # absolute value and normalized
    plt.figure()
    plt.plot(frequency/1000, np.abs(s_freq)/N)  # divide by N to normalize
                                                # frequency/1000 easier plot
    plt.xlim(-40, 40)                           # -40 to 40 kHz
    plt.xlabel("Frequency (kHz)")
    plt.ylabel("|S(f)|")
    plt.title("DSB-SC Signal Spectrum (Normalized Magnitude)")
    plt.grid()
    plt.show()
```

Question 2

Question Statement

Research/Investigate Phase-locked loops used for receivers using sources from IEEE Xplore.

Phase-Locked Loops

A phase-locked loop (PLL) is a control system designed to generate an output signal that tracks the phase of an input reference signal. When used in a receiver, this architecture acts as a high-quality bandpass filter, removing noise from the received signal to facilitate accurate data extraction.

A PLL consists of four primary components: a phase detector (PD), a loop filter (LF), a voltage-controlled oscillator (VCO), and a feedback path. The PD compares the incoming reference signal with the signal produced by the VCO. If these signals are misaligned, the PD generates an error signal. The LF processes this signal to remove high-frequency noise, outputting a stable DC control voltage. The VCO then adjusts its output frequency in response to this voltage. Finally, the feedback path returns the VCO output to the PD, closing the loop and allowing for continuous self-correction.

The following sections summarize two research articles from IEEE Xplore regarding modern PLL applications.

The first article presents a comparative analysis of Analog Phase-Locked Loops (APLL) and Digital Phase-Locked Loops (DPLL) using Simulink modeling. The research evaluated how these architectures respond to reference signals by focusing on differences in phase detector and filter design. The study concluded that the DPLL architecture significantly outperforms the APLL in terms of acquisition speed.

Figure 5 depicts the lock-in time comparison—the duration required for the PLL to synchronize its output with the reference—across various frequencies. The data illustrates that the DPLL provides a much faster response, which is essential for modern high-speed receivers. The authors argue that the DPLL's superior noise immunity and agility make it the preferred choice for digital communication, ensuring robust performance against environmental interference and reliable clock recovery.

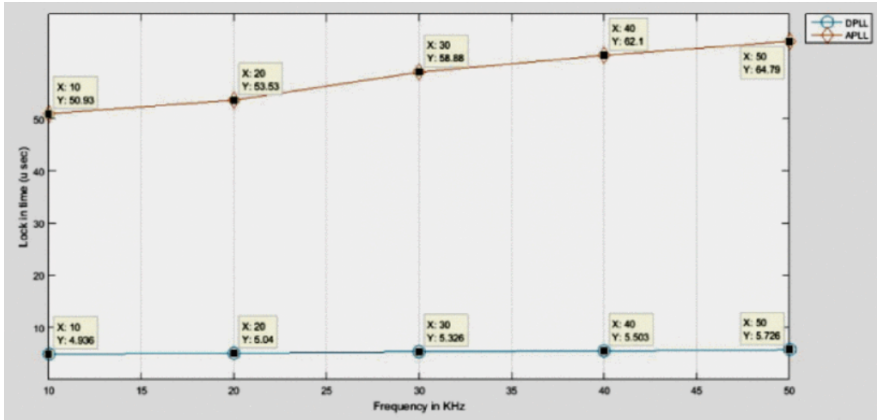

Figure 5: Lock-in time comparison between APLL and DPLL (Godave et al., 2018).

The second article addresses a critical hardware bottleneck: the VCO tuning range versus noise trade-off. Traditionally, a wide tuning range requires a high tuning gain (K_{VCO}), which inadvertently amplifies noise on the control line. The author proposes a band-switching VCO architecture that utilizes a multi-band approach. By dividing the oscillator's range into an array of smaller, overlapping bands, the system can cover a wide frequency spectrum while maintaining a low K_{VCO} within each band.

This architecture is particularly relevant for Ultra-Wideband (UWB) receivers and multi-standard devices supporting protocols such as Wi-Fi, Bluetooth, and LTE. This design allows a receiver to switch between frequency bands without the signal degradation typically associated with high-gain oscillators, maintaining spectral purity across a wide operational range.

References

[1] W.-C. Lai, "Design Band Switching Voltage Controlled Oscillator for Phase Locked Loop of Receiver Applications," in *2021 IEEE MTT-S International Microwave Workshop Series on Advanced Materials and Processes for RF and THz Applications (IMWS-AMP)*, Chongqing, China, 2021, pp. 112–114, doi: 10.1109/IMWS-AMP53428.2021.9643852.

[2] A. Godave, P. Choudhari, and A. Jadhav, "Comparison and Simulation of Analog and Digital Phase Locked Loop," in *2018 9th International Conference on Computing, Communication and Networking Technologies (ICCCNT)*, Bengaluru, India, 2018, pp. 1–4, doi: 10.1109/ICCCNT.2018.8494198.

Question 3

Question Statement

Research/Investigate Vestigial Sideband AM using sources from IEEE Xplore.

Vestigial Sideband

Vestigial Sideband (VSB) modulation is a specialized form of Amplitude Modulation (AM) that serves as a compromise between Double Sideband (DSB) and Single Sideband (SSB) techniques. In standard DSB-AM, two identical sidebands are transmitted, which is inefficient in terms of bandwidth. Both the upper and lower sidebands carry the same information. While SSB solves this by removing one sideband entirely, it requires complex filters that often distort the low-frequency components near the carrier signal. VSB resolves this by transmitting one full sideband and a small portion, or "vestige," of the other sideband.

By allowing a gradual roll-off rather than a sharp cutoff, VSB reduces the hardware complexity of the receiver while maintaining high bandwidth efficiency. VSB is historically famous for its use in television broadcasting, it is now a critical component in modern Radio over Fiber (RoF) systems, where it is used to combat signal degradation caused by optical fiber characteristics.

Below are summaries of the research conducted in two IEEE Xplore articles regarding the implementation and optimization of VSB in optical communication systems.

The first article by Ujang and Wibisono investigates a filter-less approach to generating Optical Vestigial Sideband (OVSB) signals using two parallel Dual-Drive Mach-Zehnder Modulators (DD-MZMs). The primary goal of this research was to mitigate chromatic dispersion, a phenomenon in optical fibers that causes power fading and limits transmission distance. A common method for generating a VSB signal involves passing a standard DSB signal through a precise optical filter; however, these filters typically incur a high cost due to the high precision required for their design. Furthermore, traditional fiber optic communication requires additional hardware to compensate for dispersion. By utilizing the phase and amplitude properties of the DD-MZMs, the researchers were able to suppress the unwanted sideband naturally. This creates a narrower optical spectrum that is inherently more resistant to dispersion without the need for high-cost optical filters or complex compensation hardware.

The study's results demonstrated that OVSB significantly enhances the robustness of the receiver by reducing bit error rates over long-distance transmissions, recall from the precious paragraph that dispersion causes power to face and limits transmission distance. For receiver applications, this means that data can be recovered reliably even after traveling through many kilometers of fiber. The authors conclude that this parallel modulator architecture provides a more stable and efficient signal for high-speed RoF networks than traditional DSB methods.

The second article, by Parajuli and Udvary, addresses the challenge of harmonic distortion in RoF systems through a VSB modulation scheme utilizing an MZM and a Phase Modulator (PM) in series. Harmonic distortion involves the creation of unwanted signals at frequencies that are integer multiples of the original desired signal (e.g., $2f$, $3f$). By carefully tuning the phase shift

between the two modulators, the authors successfully suppressed second-order harmonics, which typically interfere with the receiver's ability to extract the intended signal.

The authors compared the Symbol Error Rate (SER) variation of DSB, SSB, and VSB schemes across different fiber lengths. Their results indicated that the DSB scheme exhibited the highest SER as the fiber length increased, while the VSB scheme demonstrated an "improved SER compared to others." Ultimately, the study concludes that the VSB scheme most effectively addresses harmonic distortion when compared to both DSB and SSB. From a receiver design perspective, this research highlights the importance of spectral purity; the proposed architecture improves the Signal-to-Noise Ratio (SNR), ensuring that the local oscillator and demodulation stages receive a cleaner, more reliable input.

References

[1] F. Ujang and G. Wibisono, "A Vestigial Sideband Modulation Scheme Using Two Parallel DD-MZMs in Radio Over Fiber Systems," *2018 10th International Conference on Information Technology and Electrical Engineering (ICITEE)*, Bali, Indonesia, 2018, pp. 36-39, doi: 10.1109/ICITEED.2018.8534832.

[2] H. N. Parajuli and E. Udvary, "A Vestigial sideband modulation scheme in radio over fiber system using electro-optic modulators," *2016 18th International Conference on Transparent Optical Networks (ICTON)*, Trento, Italy, 2016, pp. 1-4, doi: 10.1109/ICTON.2016.7550477.